

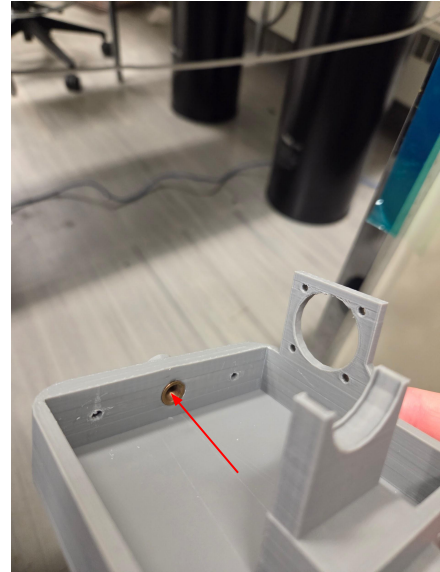
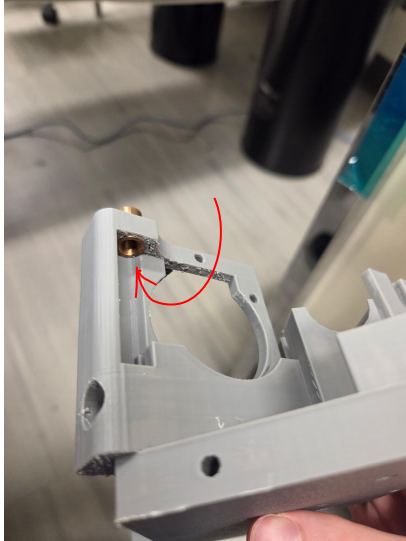
OCT Spectrometer Assembly Instructions, V1

Print 3D printed components

Our 3D printed components were printed with a Bambu P1S printer. Each component is printed with 4 layers of wall thickness at 15% sparse density infill. The base is printed with 0.08mm layer height, while the remaining parts are printed with .2 mm layer height.

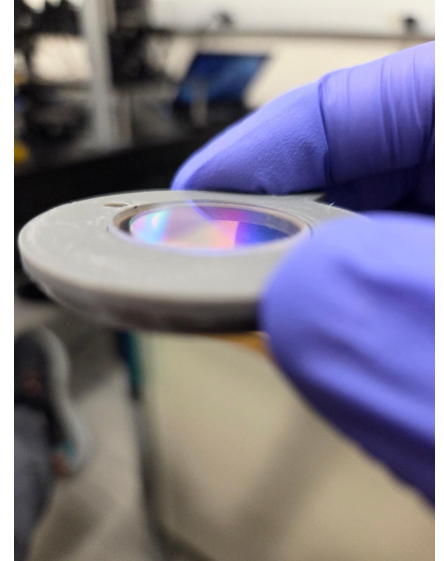
Installing the threaded inserts.

Using a heat gun to soften up the inside of the two insertion holes slightly, press fit the threaded bushing (*F25SSN2P Thorlabs*) into each hole with the lip facing inside the spectrometer.



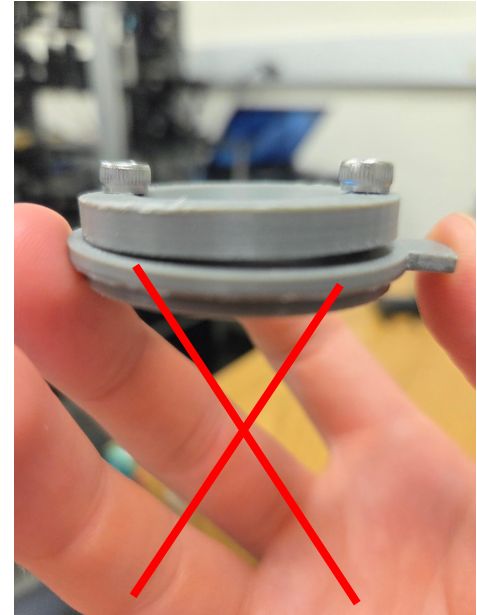
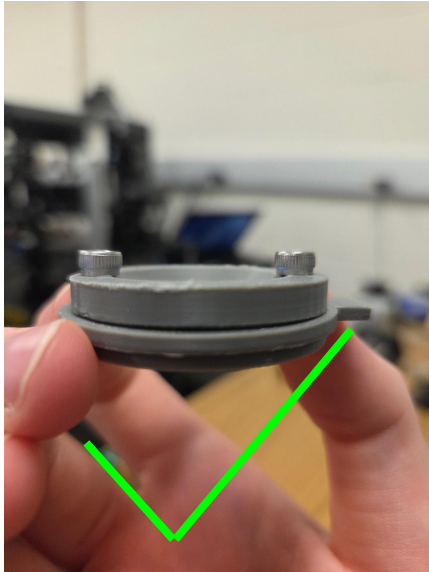
Assembling the Volume Grating Mount

Place the volume grating within the larger 3D printed holder with the vertical aligned with the two screw holes. The grating should stick out slightly. This allows the cap to screw on and hold the grating from wiggling or spinning.



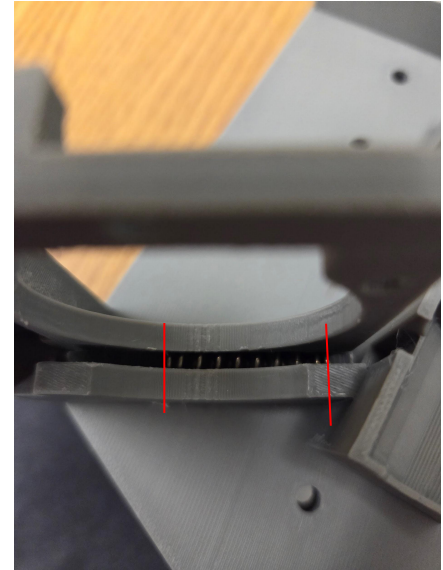
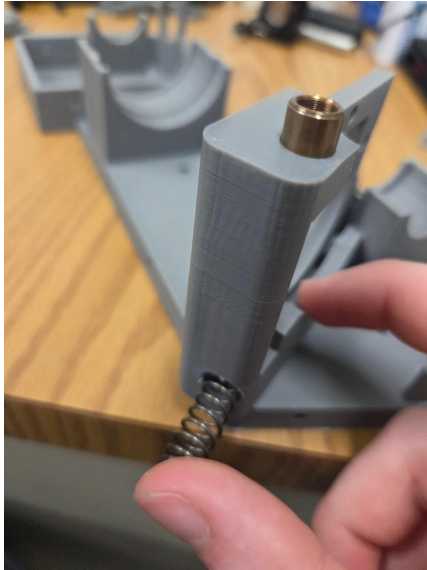
Assembling the Volume Grating Mount

Screw the cap in. Twist each screw alternating back and forth to ensure that the cap is level and not tilted to one side. The screws will thread in and out of the soft PLA plastic.



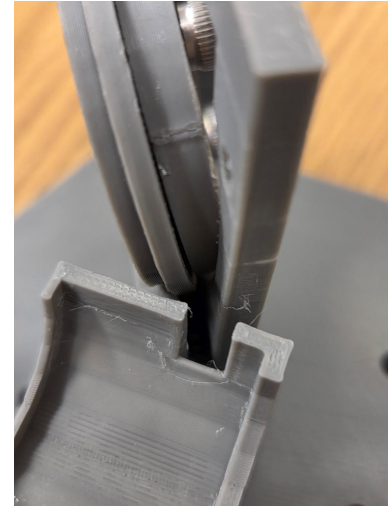
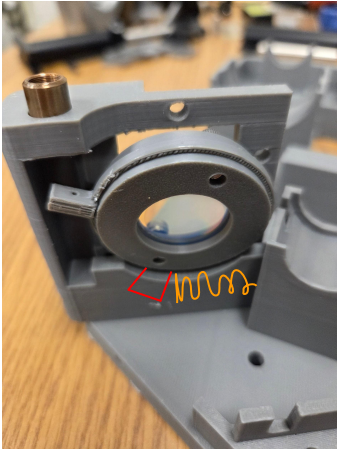
Grating Spring Insertion

Insert the spring into the grating mount and insure that it is all the way into the hole.



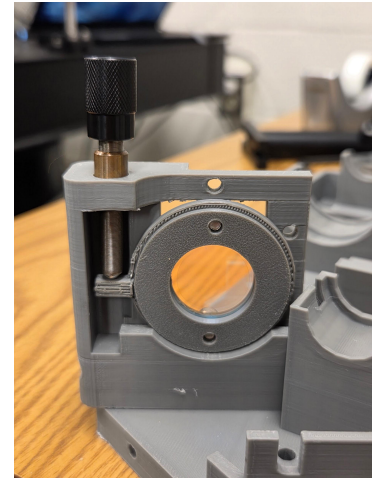
Grating Spring Insertion

Slide the grating holder into the bottom groove, ensuring that the spring is located at the side of the tooth of the grating and ensure that all of the lower sides of the grating are feeding into the groove.



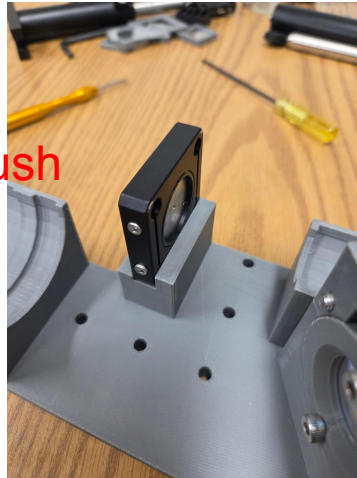
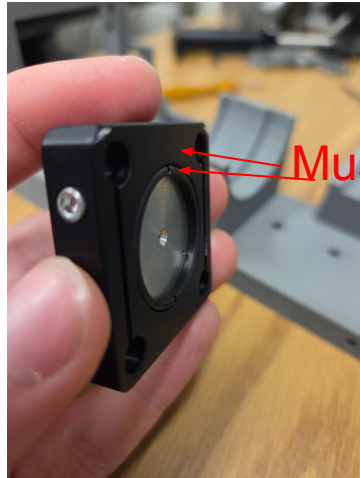
Grating Holder Insertion

Push the top of the grating further into the grating hole (may need pliers if too difficult to get in by hand).



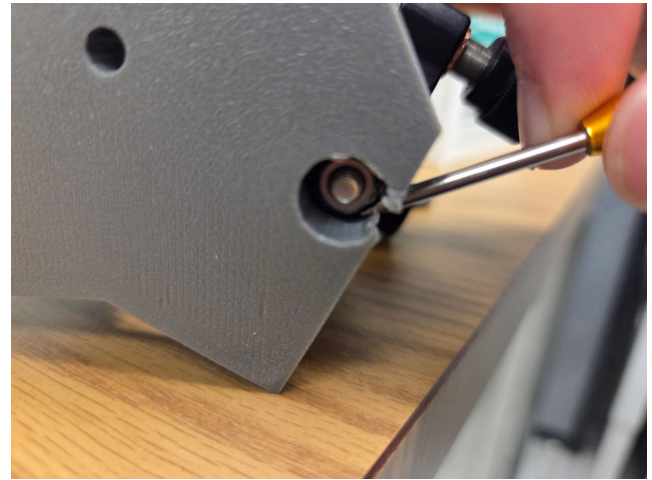
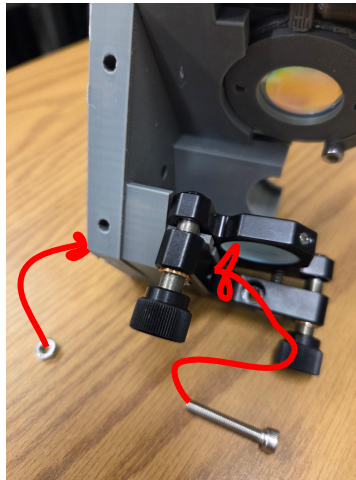
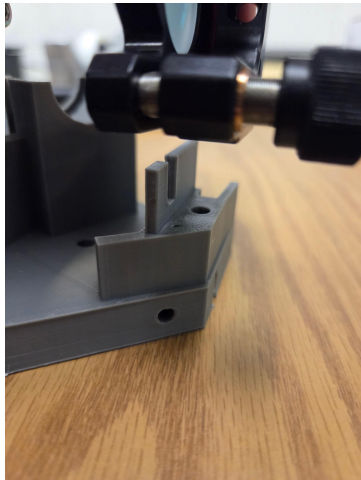
Fiber Mount Assembly

The Thorlabs fiber mount (CP08) is assembled with the provided retaining ring and the Thorlabs Fiber Adaptor Plate (SM1FCA2). Ensure that the retaining ring is screwed in until it is flush with the outside surface of the fiber mount. Position the mount with the threaded hole on one side facing the ground and push it into the holder. Screw through the bottom of the surface up into the fiber mount. It should be rigid.



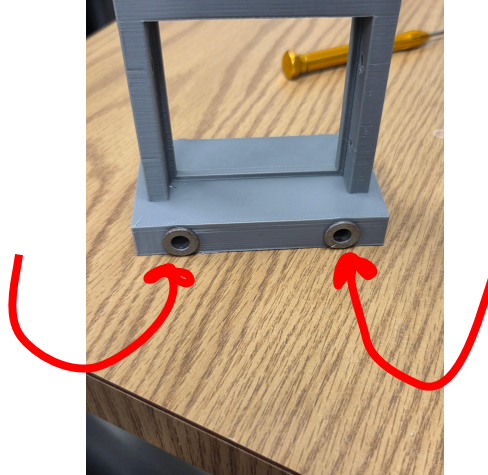
Mirror Assembly

Insert the Thorlabs Mirror (BB1-E03) into the Thorlabs mirror mount (KM100). Tighten the screw along the side of the mirror to hold it in place. Place the mirror mount along the tabs sticking out. Insert a bolt and nut in on opposite sides and tighten from the bottom. The nut free spins, so a screwdriver may be helpful to hold it in place.



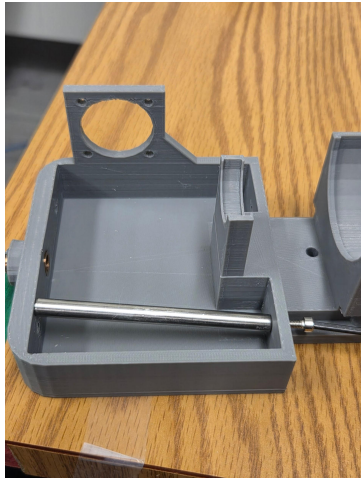
Insert bushings into trolley

Using a heat gun, soften the four plastic holes in each end of the CCD camera trolley and press fit the McMaster Carr low friction bearings (7119N117). The Thorlabs cage assembly rods (ER3) may be inserted through bearings on each end to assist with proper alignment.



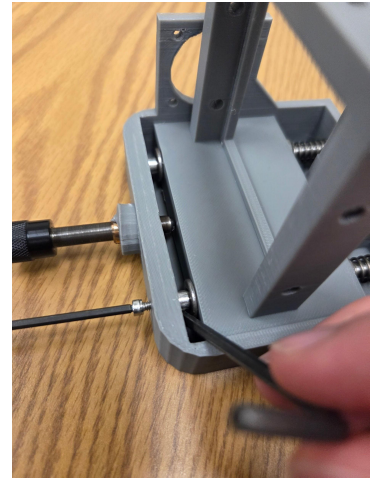
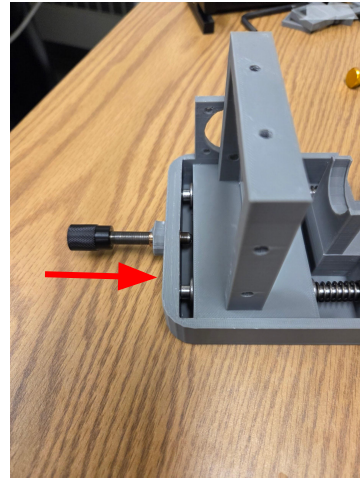
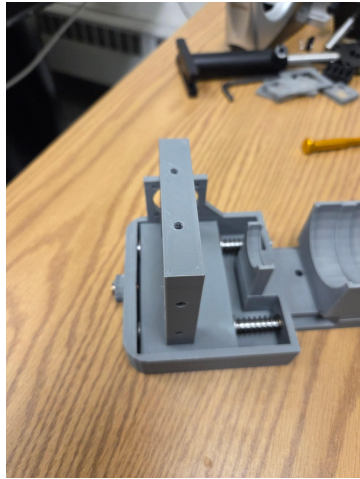
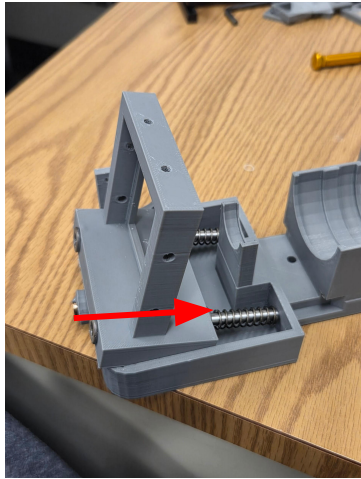
Inserting trolley rods

Screw in each of the two Thorlabs rods (ER3) through the *far side only* with the holes provided. Slide the springs onto the rods.



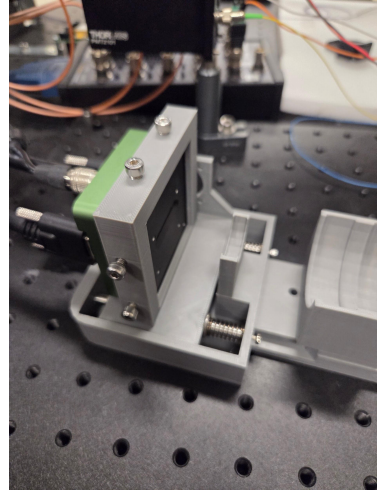
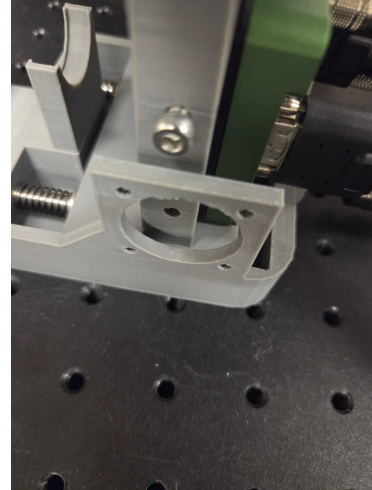
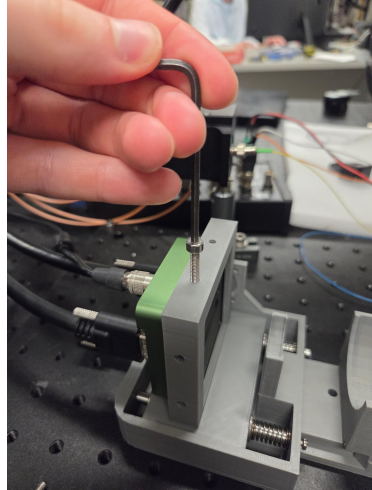
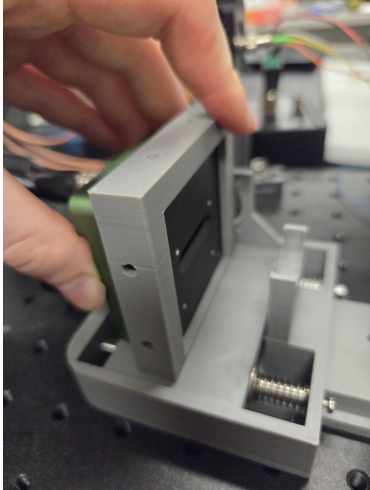
Inserting trolley rods cont.

Align the trolley bearings with the rods and push the trolley onto the rods. Insert the Thorlabs fine adjustment screw (FAS150) and screw in until the trolley is pushed forward a few millimeters. Then, align the rods with the empty holes and install the screws. The fit will be tight. The trolley can be pushed down with a finger or levered upwards with an allen wrench to properly align the rods with the hole.



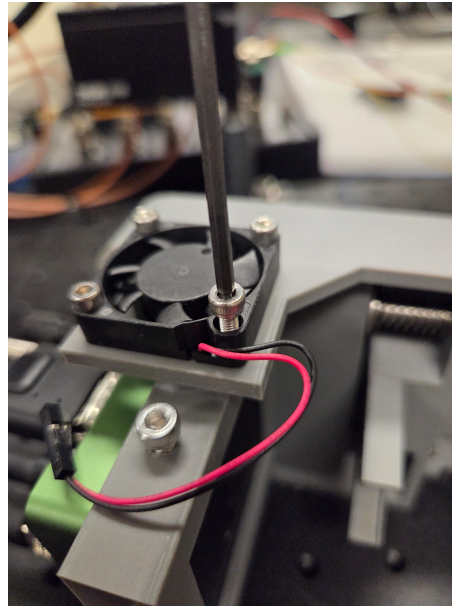
Installing CCD camera

Push the CCD camera into the trolley mount, until it is flat against the offset edge. Install the screws through each screw hole. The screws will thread into the soft plastic. If convenient, the screw that would conflict with the fan mount may be omitted. This is up to user discretion. Hook up the CCD camera to the computer and cables.



Installing the fan

Screw the Canakit fan (4468) into the fan mount. The screws will thread into the PLA. Wire as needed.



Installing the lenses

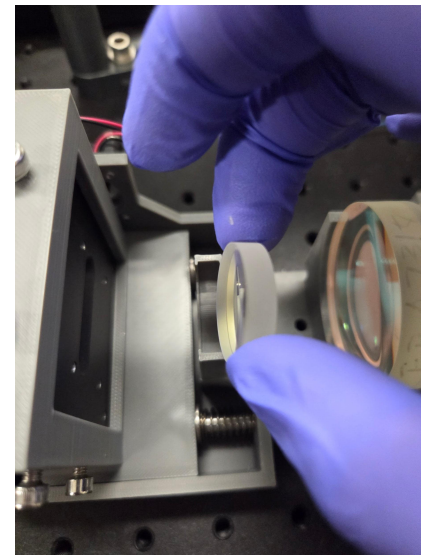
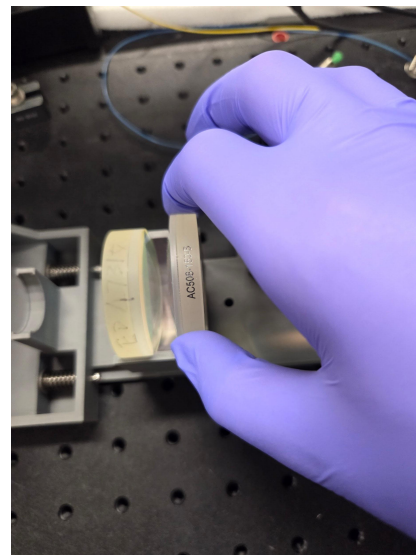
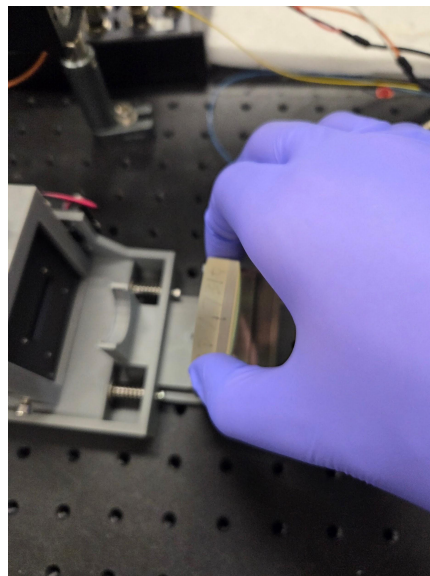
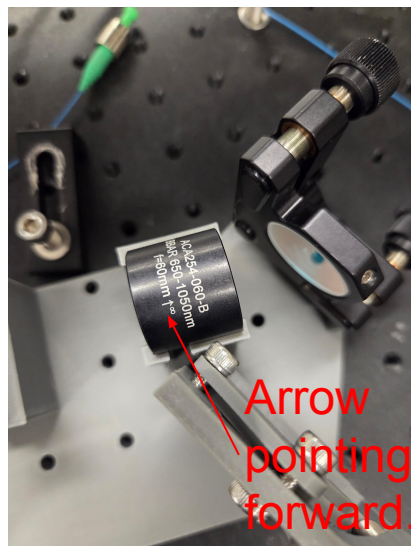
Install each of the lenses as shown below.

ACA254-060-B Thorlabs

E-47-318 Fosco

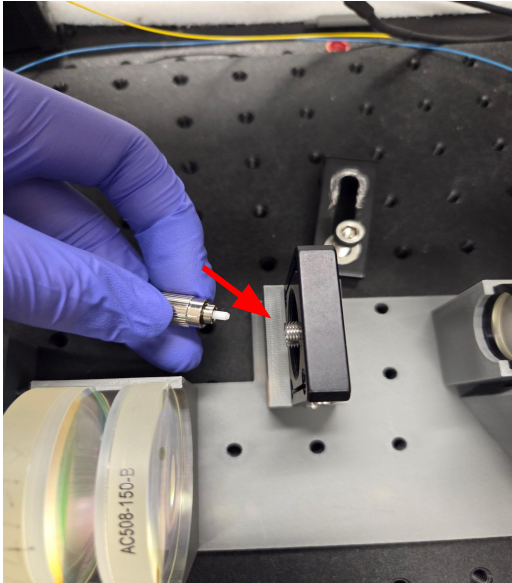
AC508-150-B Thorlabs

LC1582 Thorlabs



Installing the fiber optic

Screw the fiber optic cable into the end of the fiber optic mount. The spectrometer is now completed and ready for use.



Usage

The top knob and the bottom knob on the mirror mount are used to adjust the focused light up/down and left/right, respectively. The knob on the grating mount is used to rotate the grating to the precise degree necessary. The knob behind the CCD camera is used to move the camera back and forth in order to calibrate it into focus. If desired, the spectrometer can be clamped to a table or bolted down through the holes.

The end